Visual-field Deficits Associated with Proton Beam Irradiation for Parapapillary Choroidal Melanoma

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Purpose: A large series of patients treated with proton irradiation for parapapillary choroidal melanoma were reviewed retrospectively to determine the frequency of radiation papillopathy and visual-field loss after treatment.

Methods: Among 249 patients treated with proton irradiation for parapapillary choroidal melanoma, the authors identified 59 patients who had visual-field testing performed before treatment and at least 18 months after treatment. The visual fields, color fundus photographs, and charts were reviewed to determine the prevalence of radiation papillopathy and visual-field loss after treatment.

Results: Nineteen of the 59 patients reviewed (31%) received a clinical diagnosis of radiation papillopathy. Progressive visual-field loss, defined as enlargement of absolute scotoma of less than or equal to 30° as compared with the pretreatment visual field, was noted in 67% of patients with radiation papillopathy and 73% of patients without papillopathy. In both groups, visual-field loss correlated with the area of the retina predicted to be exposed to irradiation in the majority of patients.

Conclusions: Progressive visual-field loss is common after proton irradiation for parapapillary choroidal melanoma. However, the scotoma usually correlates with the area of the retina exposed to irradiation. The development of radiation papillopathy does not appear to be associated with additional visual-field defects in most cases.
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Various treatment modalities currently are available for the treatment of uveal melanoma. They include enucleation, external beam irradiation, and brachytherapy.[1–3] Among them, external beam irradiation with charged particles, such as protons or helium ions, administers radiation to a circumscribed region of the eye and is thought to minimize the damaging effect of radiation on neighboring normal ocular tissue.[1,4–6]

Among choroidal melanomas of the posterior pole treated with external beam irradiation or brachytherapy, radiation-induced microangiopathy can result in papillopathy and maculopathy that potentially may be visually compromising.[7–13] Despite the unfavorable location of these tumors, many patients with posterior pole melanomas prefer radiation therapy to enucleation. In the case of treatment with proton irradiation, studies have shown that most patients with choroidal melanoma near the optic disc or fovea maintain a visual acuity of 20/200 or better as long as 2 years after treatment despite the high incidence of radiation-induced papillopathy and maculopathy.[12,13]

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